Version 7.4.8

PVsyst - Simulation report

Grid-Connected System

Project: HW1_RE_810601219

Variant: New simulation variant

Sheds on ground

System power: 8.10 kWp

Tehran - Iran

Author

PVsyst V7.4.8
VC0, Simulation date:
10/25/25 15:39
with V7.4.8

Project summary

Geographical Site
Tehran
Iran

Situation

Latitude	35.69 °N
Longitude	51.42 °E
Altitude	1178 m
Time zone	UTC+3.5

Project settings

Albedo	0.20

Weather data
Tehran
Meteonorm 8.1 (1988-2000), Sat=19% - Synthetic

System summary

Grid-Connected System

Sheds on ground

PV Field Orientation
Fixed plane
Tilt/Azimuth 35 / 0 °

Near Shadings
Linear shadings : Fast (table)

User's needs
Unlimited load (grid)

System information

PV Array

Nb. of modules	18 units
Pnom total	8.10 kWp

Inverters

Nb. of units	3 units
Pnom total	9.00 kWac
Pnom ratio	0.900

Results summary

Produced Energy	14245.04 kWh/year	Specific production	1759 kWh/kWp/year	Perf. Ratio PR	83.09 %

Table of contents

Project and results summary 2
General parameters, PV Array Characteristics, System losses 3
Near shading definition - Iso-shadings diagram 4
Main results 5
Loss diagram 6
Predef. graphs 7
Single-line diagram 8

General parameters

Grid-Connected System

Sheds on ground

PV Field Orientation

Orientation	
Fixed plane	
Tilt/Azimuth	35 / 0 °

Sheds configuration	
Nb. of sheds	6 units
Sizes	
Sheds spacing	4.82 m
Collector width	2.11 m
Ground Cov. Ratio (GCR)	43.8 %
Top inactive band	0.02 m
Bottom inactive band	0.02 m
Shading limit angle	
Limit profile angle	21.7 °

Models used	
Transposition	Perez
Diffuse	Perez, Meteonorm
Circumsolar	separate

Horizon
Free Horizon

Near Shadings
Linear shadings : Fast (table)

User's needs
Unlimited load (grid)

PV Array Characteristics

PV module

Manufacturer	VSUN
Model	VSUN-450-144-MH
(Original PVsyst database)	
Unit Nom. Power	450 Wp
Number of PV modules	18 units
Nominal (STC)	8.10 kWp
Modules	6 string x 3 In series
At operating cond. (50°C)	
Pmpp	7.35 kWp
U mpp	112 V
I mpp	66 A

Inverter

Manufacturer	Huawei Technologies
Model	SUN2000-3KTL-L1
(Original PVsyst database)	
Unit Nom. Power	3.00 kWac
Number of inverters	6 * MPPT 50% 3 units
Total power	9.0 kWac
Operating voltage	80-600 V
Pnom ratio (DC:AC)	0.90
No power sharing between MPPTs	

Total PV power

Nominal (STC)	8 kWp
Total	18 modules
Module area	39.8 m²

Total inverter power

Total power	9 kWac
Number of inverters	3 units
Pnom ratio	0.90

Array losses

Thermal Loss factor

Module temperature according to irradiance	
Uc (const)	20.0 W/m²K
Uv (wind)	0.0 W/m²K/m/s

DC wiring losses

Global array res.	28 mΩ
Loss Fraction	1.5 % at STC

Module Quality Loss

Loss Fraction	-0.8 %

Module mismatch losses

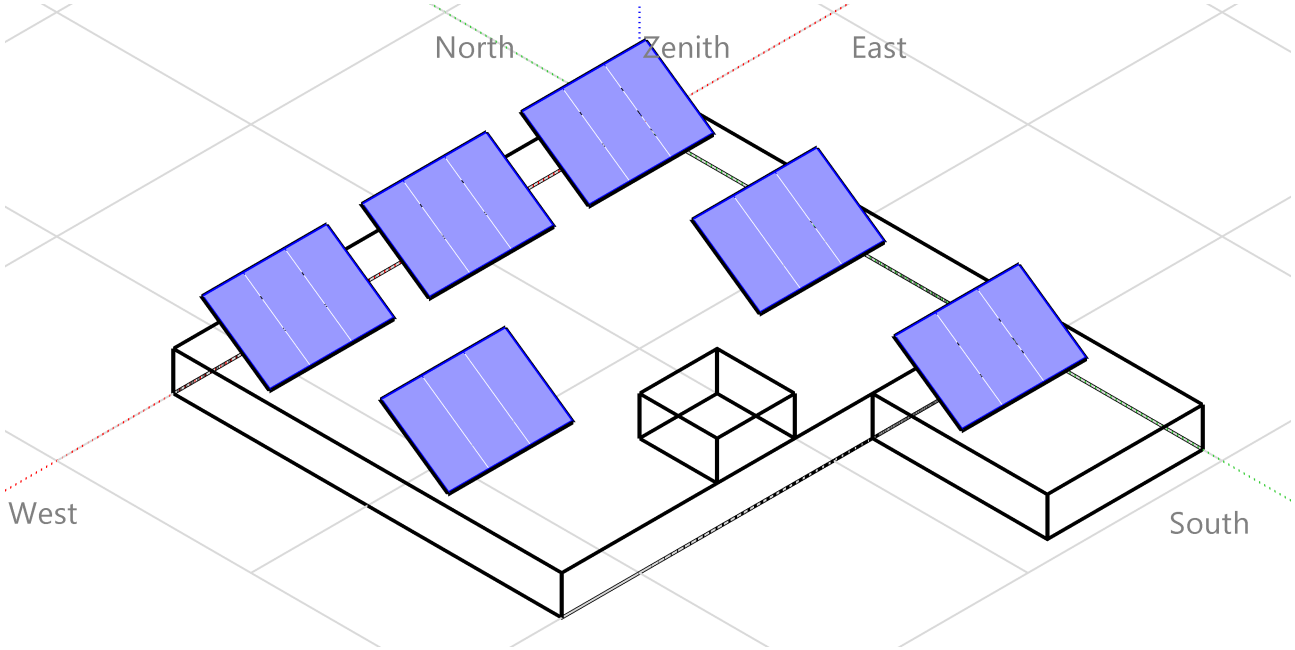
Loss Fraction	1.0 % at MPP

IAM loss factor
Incidence effect (IAM): Fresnel smooth glass, n = 1.526

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.998	0.981	0.948	0.862	0.776	0.636	0.403	0.000

Near shadings parameter

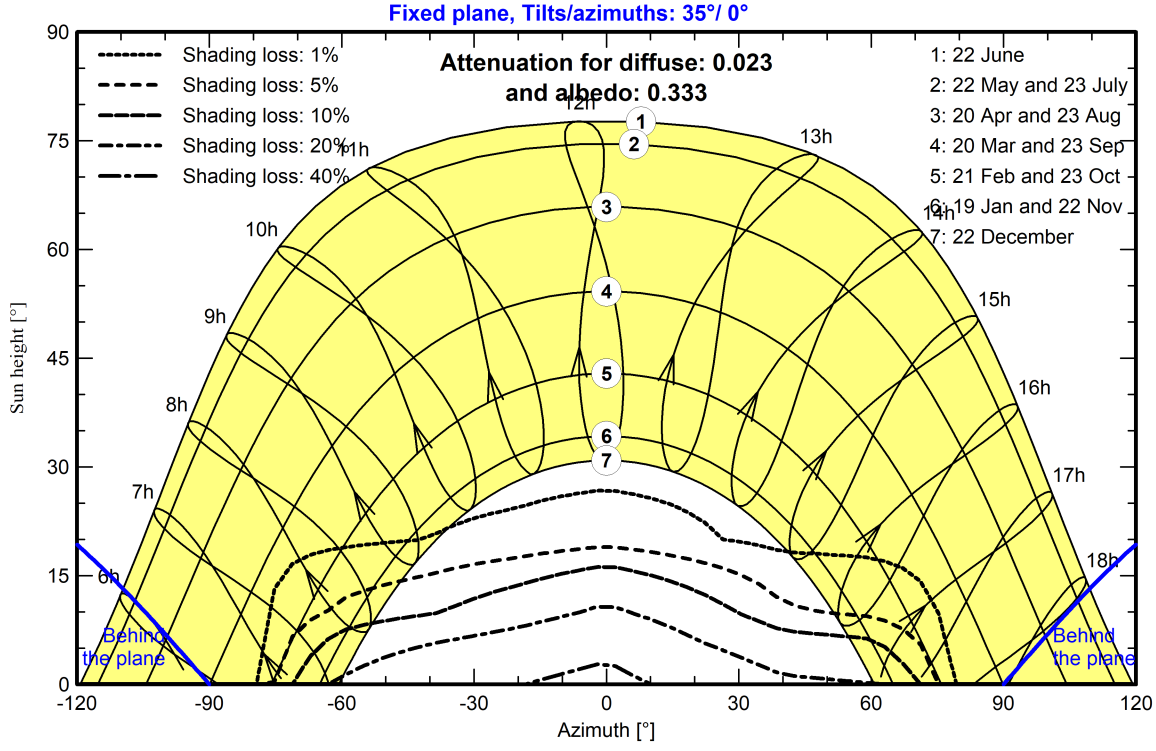
Perspective of the PV-field and surrounding shading scene



Iso-shadings diagram

Orientation #1

Fixed plane, Tilts/azimuths: 35°/ 0°

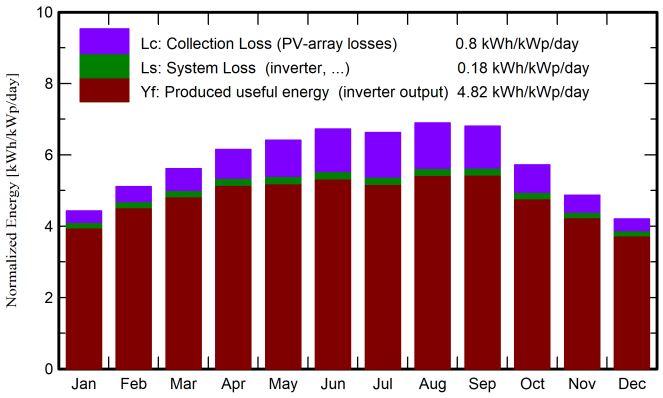


Main results

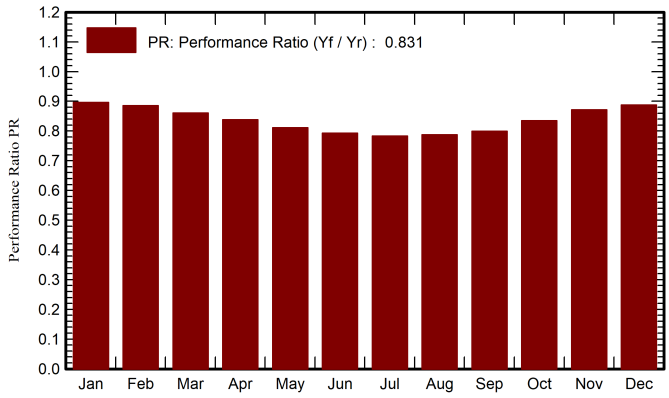
System Production

Produced Energy	14245.04 kWh/year	Specific production	1759 kWh/kWp/year
		Perf. Ratio PR	83.09 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

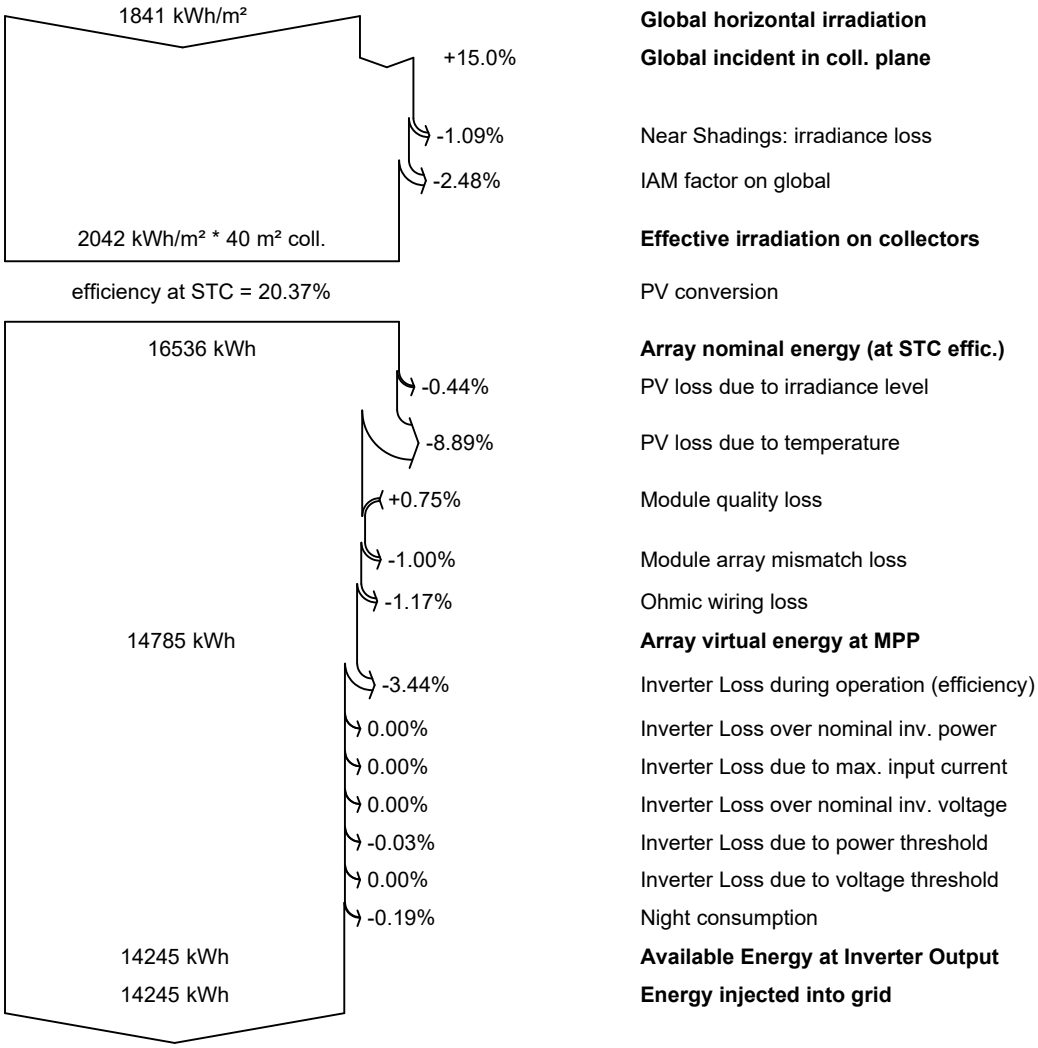
	GlobHor kWh/m²	DiffHor kWh/m²	T_Amb °C	GlobInc kWh/m²	GlobEff kWh/m²	EArray kWh	E_Grid kWh	PR ratio
January	83.3	29.83	5.08	137.1	132.7	1033	995	0.896
February	99.6	34.42	7.26	143.1	139.0	1065	1026	0.886
March	142.5	54.73	12.75	174.0	168.3	1258	1213	0.860
April	174.2	65.79	17.59	184.6	177.6	1299	1252	0.838
May	209.5	70.09	23.73	198.7	190.5	1356	1305	0.811
June	224.7	73.09	28.75	201.8	193.0	1346	1296	0.793
July	222.3	77.43	31.66	205.3	196.7	1353	1302	0.783
August	209.7	65.80	30.66	213.8	205.8	1414	1363	0.787
September	174.8	45.45	26.24	204.3	197.4	1371	1323	0.800
October	131.8	43.74	19.97	177.4	172.5	1244	1199	0.835
November	91.8	31.48	11.30	146.1	141.4	1070	1031	0.872
December	76.5	26.05	6.44	130.4	126.5	975	938	0.888
Year	1840.6	617.91	18.52	2116.6	2041.5	14785	14245	0.831

Legends

GlobHor	Global horizontal irradiation
DiffHor	Horizontal diffuse irradiation
T_Amb	Ambient Temperature
GlobInc	Global incident in coll. plane
GlobEff	Effective Global, corr. for IAM and shadings
EArray	Effective energy at the output of the array
E_Grid	Energy injected into grid
PR	Performance Ratio

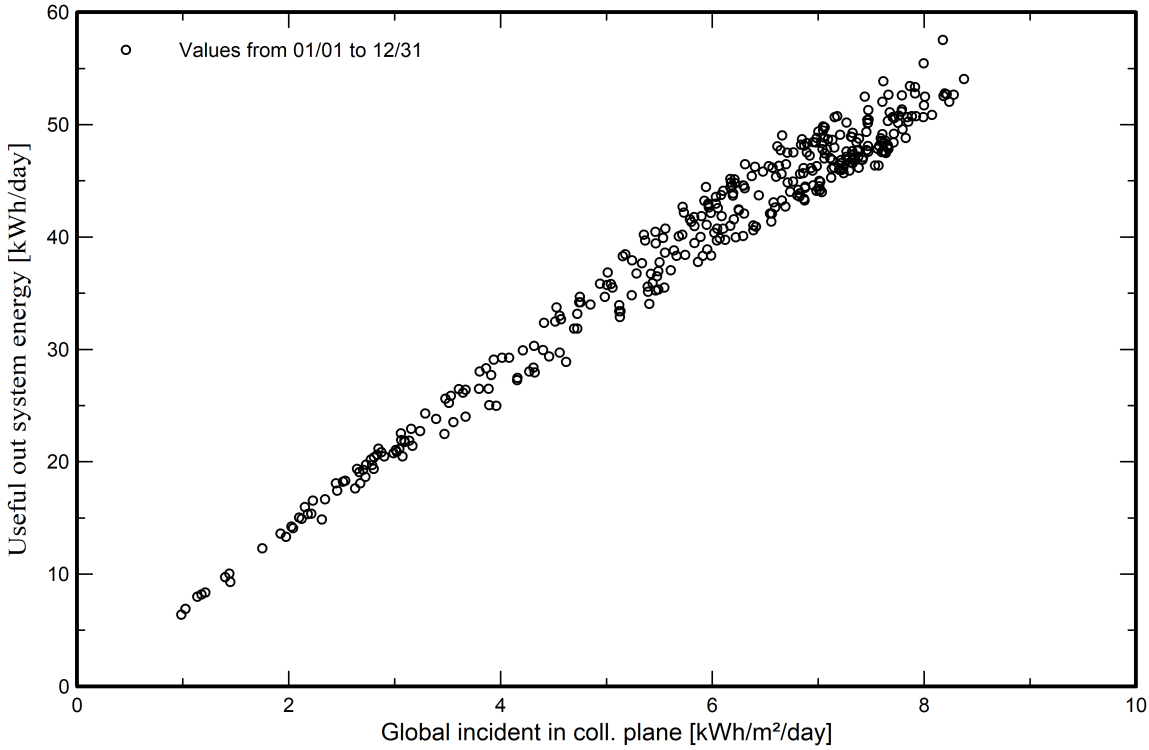
PVsyst V7.4.8
VC0, Simulation date:
10/25/25 15:39
with V7.4.8

Loss diagram

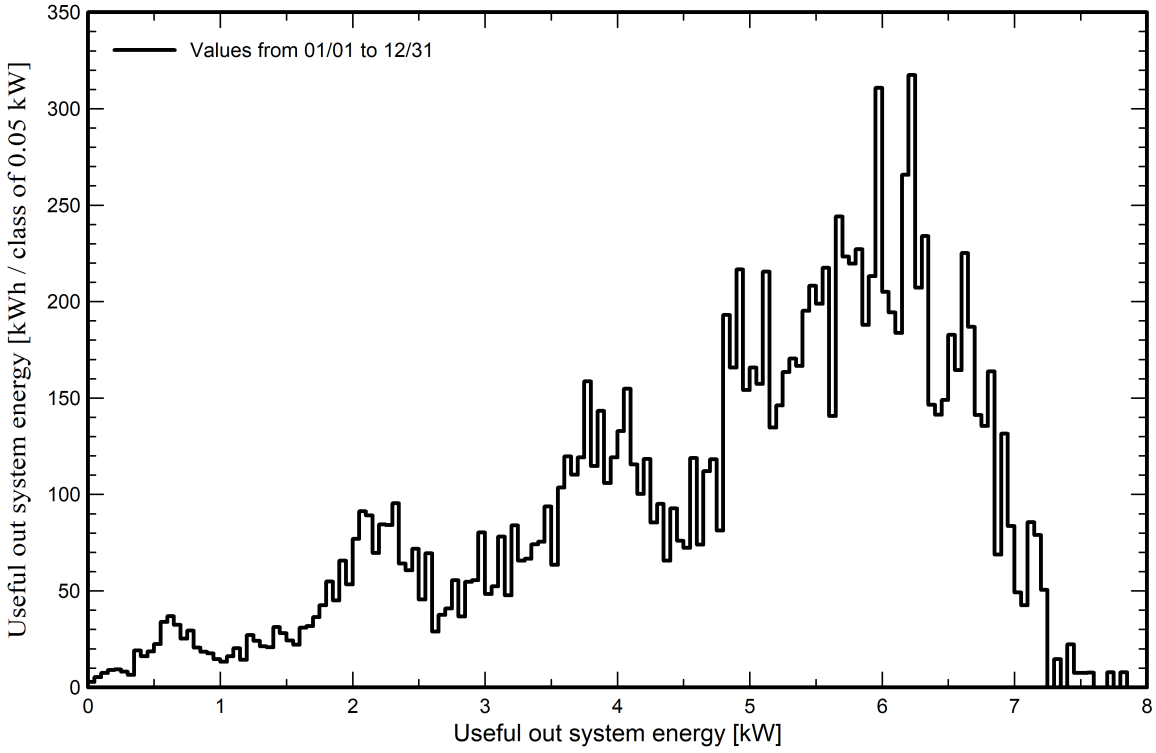


Predef. graphs

Daily Input/Output diagram

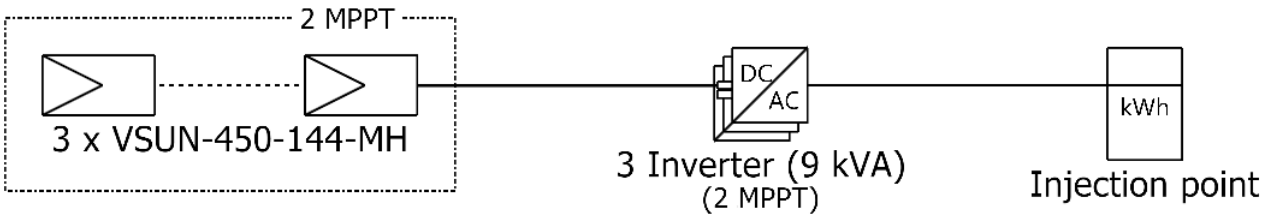


System Output Power Distribution



Single-line diagram

PVsyst V7.4.8
VC0, Simulation date:
10/25/25 15:39
with V7.4.8



PV module	VSUN-450-144-MH
Inverter	SUN2000-3KTL-L1
String	3 x VSUN-450-144-MH

	HW1_RE_810601219	
	VC0 : New simulation variant	10/25/25